

CA Note

1 Basic Knowledge

Group $(G, *)$: **G1**: Closure **G2**: Associativity **G3**: Identity element **G4**: Inverse element

Ring $(R, +, \cdot)$: **R1**: $(R, +)$ is an abelian group **R2**: $(R, \cdot)$ is monoid. (no inverse) **R3**: Distributive law (left and right)

Ideal (I) : **I1**: $I \neq \emptyset$ **I2**: $\forall a, b \in I, a - b \in I$ **I3**: $\forall r \in R, i \in I, ri, ir \in I$ **Lemma**: $I = R \Leftrightarrow 1 \in I \Leftrightarrow I \cap R^\times \neq \emptyset$

Subgroup (H) : **H1**: $e \in H$ **H2**: $\forall a, b \in H, ab \in H$ **H3**: $\forall a \in H, a^{-1} \in H$ **Subring** (S) : **S1**: $1 \in S$ **S2**: $\forall a, b \in S, a - b \in S$ **S3**: $\forall a, b \in S, ab \in S$

Integral Domain (ID): **ID1**: commutative ring **ID2**: no zero divisors **Lemma**: Finite ID $\Rightarrow$ field ID $\Rightarrow$ Cancellation Law

Factor Ring (R/I) : $a + I = b + I \Leftrightarrow a - b \in I$ $(R/I, \oplus, \odot)$ is a ring. ($1_{R/I} = 1_R + I, 0_{R/I} = 0_R + I, I$ ideal)

Ring Homomorphism: $\varphi: R \rightarrow S$ s.t. **RH1**: $\varphi(a + b) = \varphi(a) + \varphi(b)$ **RH2**: $\varphi(ab) = \varphi(a)\varphi(b)$ **RH3**: $\varphi(1_R) = 1_S$

• **1**: $\text{Ker}(\varphi)$ is an ideal. **2**: $\text{Im}(\varphi)$ is a subring. **3**: φ is 1-1 $\Leftrightarrow \text{Ker}(\varphi) = \{0_R\}$. **FIT**: $R/\text{Ker}(\varphi) \cong \text{Im}(\varphi)$ (First Isomorphism Theorem)

• **Canonical Homomorphism**: $\text{can}: R \rightarrow R/I$ s.t. $\text{can}(r) = r + I$ is a onto ring Homomorphism. $\text{Ker}(\text{can}) = I \quad \updownarrow (\exists! \tilde{\varphi} \text{ 双射})$

• **Lemma**: If $\varphi: R \rightarrow S$ ring Homomorphism s.t. $I \subseteq \text{Ker}(\varphi)$, then $\exists! \tilde{\varphi}: R/I \rightarrow S$ s.t. $\varphi = \tilde{\varphi} \circ \text{can}$ (i.e. $\tilde{\varphi}(r + I) = \varphi(r)$)

Lagrange's Theorem: G finite group, $H \leq G$ subgroup, then $|H| \mid |G|$

2 Types of Ideals

Division: If R ring, then: $\forall f, g \in R[x], g$ 首相系数可逆 $\Rightarrow \exists q, r \in R[x]$ s.t. $f(x) = g(x)q(x) + r(x)$ and $\deg(r) < \deg(g)$ or $r = 0$.

Lemma: If R ring, $g \in R[x], g \neq 0, I = (g)$. $\Rightarrow \forall i \in R[x]/I, i = f + I, \deg(f) < \deg(g)$

g 首相系数可逆 $\Rightarrow$ If $a, b \in R[x], \deg(a, b) < \deg(g)$ and $a + I = b + I$, then $a = b$. (系数也相同)

Remark: 有些时候我们能用 gcd, lcm 计算 $\langle a, b \rangle = \langle \gcd(a, b) \rangle$ 和 $a \cap b = \langle \text{lcm}(a, b) \rangle$ [If R commutative ring] 但注意! 要在能够用 Division Algorithm | 有 gcd, lcm 的环里用.

Irreducible: If R in ID, $0 \neq a \in R$ (not unit) is irreducible $\Leftrightarrow a \neq bc$ for any non-units $b, c \in R$

Principal Ideal: Ideal I : iff $I = aR$ **Prime Ideal**: Ideal $I \neq R$: $ab \in I \Rightarrow a \in I$ or $b \in I$. (OR: If $a \notin I$ and $b \notin I$, then $ab \notin I$)

Maximal Ideal: Ideal $I \neq R$: If J ideal and $I \subseteq J \subseteq R$, then $J = I$ or $J = R$.

Radical|**Radical Ideal**: I : **Radical**: $\sqrt{I} = \{r \in R : r^n \in I \text{ for some } n \in \mathbb{N}\}$ **Radical Ideal**: If $I = \sqrt{I}$ $I \subseteq \sqrt{I}$ always.

Properties: Relationship between these kinds of ideals: **Maximal Ideals** $\subseteq$ **Prime Ideals** $\subseteq$ **Radical Ideals** $\subseteq$ **Ideals**

1. R commutative ring, I ideal. **I prime** $\Leftrightarrow R/I$ is an ID. **I maximal** $\Leftrightarrow R/I$ is a field **I maximal** $\Rightarrow$ **I prime**

2. R ring, I ideal, J ideal of $R/I, T = \{r \in R : r + I \in J\}$ (reps of J). Then **1** T : **Ideal of R** **2** $I \subseteq T$ **3** $J = \{t + I : t \in T\}$ **4** $(R/I)/J \cong R/T$

3. R commutative ring, I ideal. **I radical** $\Leftrightarrow R/I$ has no nilpotent elements except 0 **I prime** $\Rightarrow$ **I radical** (nilpotent: $r^n = 0$)

4. If R (commutative with unity), then any ideal $I \neq R$ is contained in a maximal ideal. If r is not a unit, then $1 \notin \langle r \rangle = rR$

3 Polynomial Rings

Polynomial Ring: R commutative ring. $R[x_1, \dots, x_n]$ is a ring. The elements of it: $\sum_{\alpha} a_{\alpha} x^{\alpha}$ $\alpha = (\alpha_1, \dots, \alpha_n), a \in R, x^{\alpha} = x_1^{\alpha_1} \dots x_n^{\alpha_n}$

- **Degree**: $\deg(f) = \max\{\sum_i \alpha_i : a_{\alpha} \neq 0\}$ If $f = 0, \deg(f) = -\infty$. **2nd Definition**: $R[x, y] = R[x][y]$ and so on. (induction 喜欢)
- $\deg(fg) \leq \deg(f) + \deg(g)$ unless R is an ID. If R is an ID, then $R[x_1, \dots, x_n]$ is an ID.